

ment positions, including job titles, start and end dates, and the employing firm.¹⁶

A key feature of the dataset is the standardized *seniority level* variable for each position, constructed by Revelio through an ensemble modeling approach based on multiple sources of information. This measure combines information from (i) the worker’s current job (title, firm, and industry), (ii) their work history (tenure and previous seniority), and (iii) their age. These three inputs produce separate scores, which are averaged and then categorized into seven standardized seniority levels: Entry Level, Junior Level, Associate Level, Manager Level, Director Level, Executive Level, and Senior Executive Level. In the analysis that follows, we group positions into two broad categories: *juniors* (Entry and Junior) and *seniors* (Associate and above).¹⁷ Supplemental Appendices A.1 and A.2 provide comprehensive data validation; Section 4.3 below summarizes their main findings.

We merge the worker résumé data with Revelio’s job postings database at the firm level. The job postings database tracks firms’ recruitment activity since September 2021 and contains, for each posting, a firm identifier, posting date, and the raw text of the job description. We use these job-posting texts to construct our measure of firm-level GenAI adoption, as described in Section 4.2.

In addition, we incorporate the occupational GenAI exposure measures from Eloundou et al. (2024), merged with our position-level data via O*NET SOC codes. We rely on the GPT-4–based beta exposure measure (as in, for example, Brynjolfsson et al., 2025a), and classify all the positions into three categories: low exposure (0–25th percentile), medium exposure (25–75th percentile), and high exposure (75–100th percentile).

We also construct a representative sample of *tasks* from Revelio’s job postings database. The extraction pipeline is motivated by Hampole et al. (2025), who construct firm-level AI exposure by mapping résumé text to O*NET tasks. From the Revelio job-postings corpus we draw a representative sample of 1,000 firms containing 355,013 postings, and map each posting’s free-text description to a subset of the 19,265 standardized O*NET tasks using a two-stage pipeline. The first stage is a TF-IDF retriever that returns the top 50 candidate O*NET tasks for each posting; the second stage is a large language model

¹⁶The initial data download was performed in May 2025. In April 2026, we re-downloaded the positions data to capture profile updates that users added with a lag, ensuring more complete coverage of recent employment spells.

¹⁷More details on Revelio’s seniority classification methodology are available at <https://www.data-dictionary.reveliolabs.com/methodology.html#seniority>.

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

0.56 years for juniors and 0.28 years for seniors at 2022Q4—remains roughly constant throughout the sample, with no visible break around the ChatGPT launch. The upward trend in each series reflects the gradual aging of the observed workforce: average LME rises by approximately 2.8 years in all four series over the five-year window.

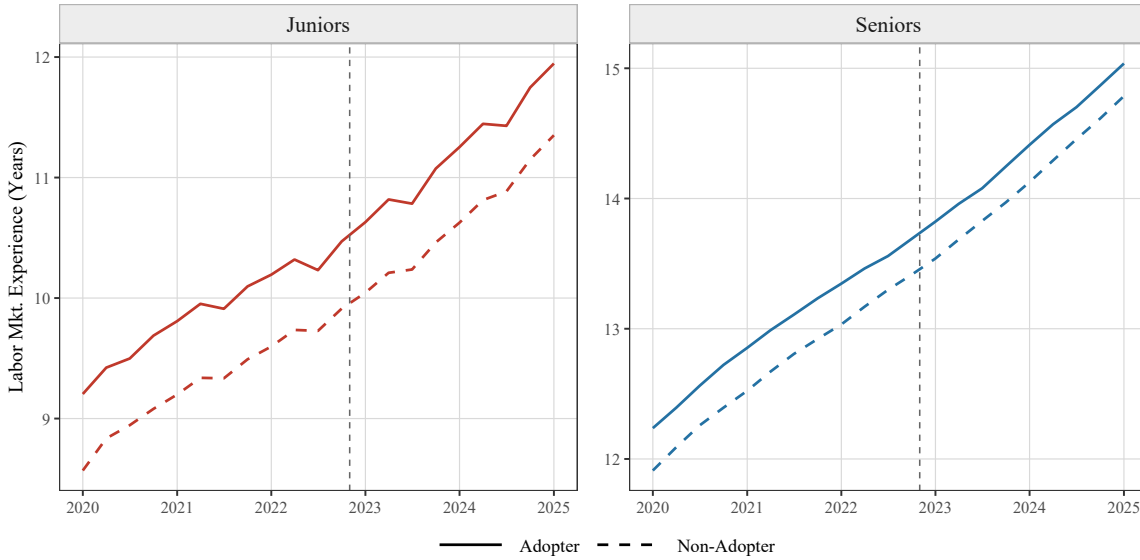


Figure A.10: Average Labor Market Experience by Seniority and GenAI Adoption Status

Notes: This figure plots average Labor Market Experience (LME), defined as years since the worker’s first observed position, for juniors (left panel) and seniors (right panel) at GenAI-adopting (solid) and non-adopting (dashed) firms over time. Averages are worker-weighted within each firm-quarter cell and then aggregated across firms by adoption status. The vertical dashed line marks the launch of ChatGPT in November 2022. Sample: quarters 2020Q1–2025Q1.

To isolate differential dynamics from the common time trend, we estimate, separately for juniors and seniors, the specification

$$\overline{\text{LME}}_{f,t} = \alpha_f + \alpha_t + \sum_{k \neq 2022Q4} \beta_k \cdot \mathbf{1}\{t = k\} \cdot \text{Adopter}_f + \varepsilon_{f,t},$$

where f indexes firms and t indexes quarters. The regression is weighted by the number of workers in each firm-quarter cell, and standard errors are clustered at the firm level. Each coefficient β_k captures the adopter–non-adopter difference in average LME at quarter k , relative to the baseline quarter 2022Q4.

Figure A.11 reports the results. For juniors, the coefficients are tightly centered around zero throughout the sample, with no discernible break at the ChatGPT launch. For se-

A.10 College Graduates Unemployment

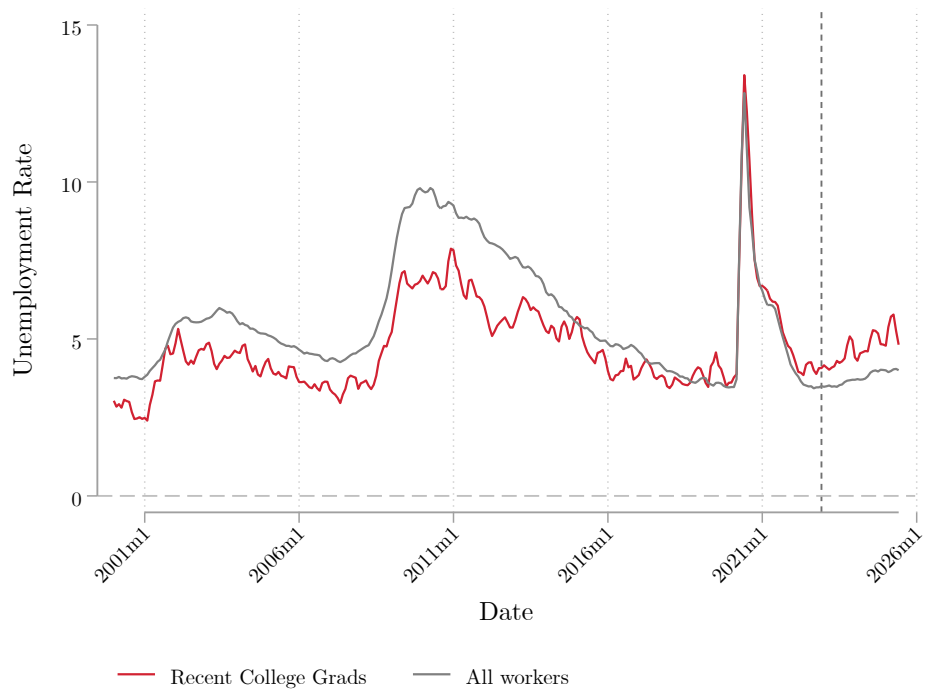


Figure A.18: Unemployment Rates for Recent College Graduates vs. All Workers

Notes: Source: Federal Reserve Bank of New York, *Labor Market for Recent College Graduates*. The red line shows the unemployment rate for recent U.S. college graduates (aged 22–27 with a bachelor’s degree), and the blue line shows the unemployment rate for all U.S. workers, on a monthly basis. Since late 2022, the college-graduate rate has risen even as the overall young worker rate remained mostly flat.